Week: 8 (19-23 October 2026)

الأسبوع #: من - إلى

Subject/الموضوع: Advanced Physics **Unit Titleعنوان الوحدة: Rotational Motion Graphs & Assessment** **Grade12: الصف**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 الحصة 1: –Angular Position Time Graphs	• I can interpret angular position–time (θ–t) graphs and calculate angular velocity from the gradient of a straight segment.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Rotational graph notes and practice on Teams; calculator and ruler.
Lesson 2 الحصة الثانية Angular Velocity–Time Graphs	• I can find angular acceleration from the gradient and angular displacement from the signed area under angular velocity–time (ω–t) graphs.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Graph worksheet and class examples on Teams; calculator and ruler.
Lesson 3 الحصة الثالثة Summative Revision	• I can review the shared assessment scope and apply rotational motion concepts, equations and graph skills to solve problems.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Assessment scope, revision sheets and previous formative resources on Toddle/Teams.
Lesson 4 الحصة الرابعة Summative Assessment	• I can demonstrate conceptual understanding, mathematical application and graph interpretation in the summative assessment.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. Summative: individual written assessment. Approved access arrangements applied where required.	Revise the shared assessment scope; bring calculator, ruler and required stationery.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Bring a calculator, ruler, notebook and stationery. Parents: encourage revision of graph gradients, signed areas and the shared assessment scope; check the assessment date.		

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Technology Integration دمج التكنولوجيا	Teams: graph notes and practice resources. Toddle: formative resources and assessment information. Digital graphing may support comparing motion representations.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: gradients, areas of triangles and trapezia, graph scales and signed quantities. Engineering: interpreting rotating machinery data.
External Assessment Integration دمج التقييمات الخارجية	Practise interpreting motion graphs, selecting equations and explaining solutions in structured assessment questions.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: connect graph features to rotational motion. Communication: explain gradients, areas and signs. Self-management: organise revision and check units. Wellbeing: build confidence through supported practice.
Standards المعايير	• Rotational kinematics: use graphical and mathematical representations to analyse angular position, angular velocity, angular acceleration and angular displacement.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can a graph reveal whether a rotating object speeds up, slows down or changes direction?
Success Criteria معايير النجاح **(Measurable)**	MUST (DOK 1): Identify graph axes and units; describe rest, rotation direction and changes in angular velocity. SHOULD (DOK 2): Calculate angular velocity from θ–t gradients, angular acceleration from ω–t gradients and angular displacement from signed areas. COULD (DOK 3): Analyse multi-stage rotational graphs, justify any reversal and compare total angular displacement with angular distance. Revision: select methods and explain solutions. Summative: demonstrate the assessed outcomes independently.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding التفاهم العالمي Arabic Language اللغة العربية
	Area: Innovation and Infrastructure المجال
	Implementation: Use a model of an observation wheel in Dubai or rotating machinery to discuss how engineers interpret التطبيق rotational motion graphs to plan smooth starts and stops and support safe operation.